

Fig. 5: Actual-size PCB layout for transmitter section

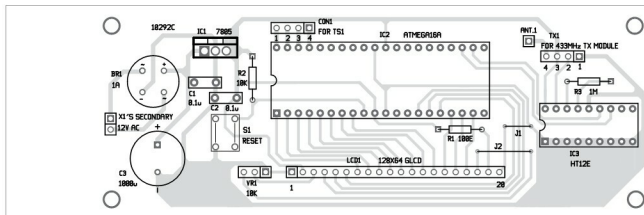


Fig. 6: Components layout for the PCB (transmitter section)

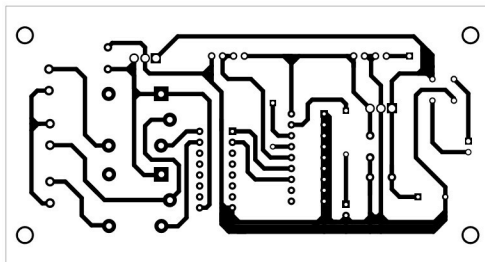


Fig. 7: Actual-size PCB layout for receiver section

around transformer X2, bridge rectifier BR2, 5V voltage regulator 7805 (IC4), decoder HT12D (IC5), 433MHz RF receiver module (RX1), an antenna (ANT.2), relay driver ULN2003 (IC6), 12V voltage regulator 7812 (IC7), two 12V, single-changeover relays (RL1 and RL2) and a few other components.

In the receiver section, data pins

of RX1 are connected to DIN (pin 14) of decoder HT12D (IC5). All address pins of IC5 are connected to ground as in IC3 in transmitter section. Two output pins (D10 and D11) of IC5 are connected to two input pins IN1 (pin 1) and IN2 (pin 2) of ULN 2003 (IC6). Output pins 16 and 15 of IC6 are connected to relays RL1 and RL2, respectively. Two voltage regulators (5V and

12V) are used here, 5V for the circuit and 12V for the relays.

Working of the receiver section is simple. When the user touches 'light on' symbol on the touchscreen, data in the form of 1s and 0s is transmitted through ANT.1 and reaches the receiver through ANT. 2. This data reaches decoder IC5 through 433MHz RF Rx module. IC5 decodes the data and output to IC6. Respective relays are energised through IC6. The appliances are switched on through the contacts of the respective relays.

If the user touches 'light off' symbol on the touchscreen, D10 output of IC5 becomes low. This low output is applied to pin 2 (IN2) of IC6, causing pin 15 (OUT2) to become high. This de-energises relay RL2, and the bulb is turned off. Similar is the case with the fan.

Software

The code is written in C programming language for the AVR and compiled in Atmel Studio 6.2. The program